

Dr. Varsha Singh

AI Researcher | Computer Vision | Generative AI | Human-Centered Computing

DST WISE-SCOPE Postdoctoral Fellow

Indian Institute of Information Technology Allahabad

Email: varshasingh.aiml@gmail.com

GitHub: <https://github.com/varsha309>

Google Scholar: <https://scholar.google.com/citations?user=ZsEut5QAAAAJ&hl=en>

LinkedIn: <https://www.linkedin.com/in/varsha-singh-b1b882b1/>

ORCID: <https://orcid.org/0000-0003-4777-1278>

Profile

AI researcher with over 13 years of teaching experience and expertise in Artificial Intelligence, Computer Vision, Deep Learning, Generative AI, Multimodal Learning, and Human-Centered Computing. Research focuses on intelligent systems for healthcare, accessibility, education, and assistive technologies.

Research Interests

Artificial Intelligence • Machine Learning • Deep Learning • Computer Vision • Generative AI
• Multimodal Learning • Medical Image Analysis • Human-Centered AI • Assistive Technologies
• Image Captioning • Large Language Models • Cognitive Computing

Education

Ph.D. (Information Technology), IIIT Allahabad, 2024

Thesis: Attention Based Models for Generating Textual Description from Images and Visual Content from Text

M.Tech. (Computer Science & Engineering), UPTU (AKTU), 2011

B.Tech. (Computer Science & Engineering), UPTU (AKTU), 2005

Experience

DST WISE-SCOPE Postdoctoral Fellow, IIIT Allahabad (2025–Present)

Temporary Faculty, MNNIT Allahabad (2024–2025)

Assistant Professor, JSS Academy of Technical Education (2007–2018)

Lecturer, NIET (2006–2007)

Selected Publications

- MAT2I: Enhancing Perceptual Authenticity in Text-to-Image Synthesis Using Multi-Attribute GANs (JAIR, 2025)
- Scene Description with Context Information Using Dense-LSTM (2023)
- Visual Content Generation from Textual Description Using Improved Adversarial Networks (MTAP, 2023)
- Indian Sign Language Recognition Using SURF with SVM and CNN (Array, 2022)

Current Projects

- DST WISE-SCOPE Project on AI-assisted communication and cognitive development for children with hearing impairment.
- Deep Learning for Facial Emotion Recognition.
- Attention-based Image Captioning.

Awards & Achievements

- DST WISE-SCOPE Fellowship
- MHRD Research Fellowship
- SERB-DST Travel Grant
- Best Paper Award (FICTA 2023)
- UGC-NET Qualified
- GATE (CSE) Qualified
- IEEE Reviewer
- Session Chair & Session Organizer (IHCI)
- Distinguished Academic Speaker

Teaching Areas

Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Data Structures, Compiler Design, Design and Analysis of Algorithms, DBMS, Python Programming

Technical Skills

Python, C/C++, MATLAB, PyTorch, TensorFlow, Keras, OpenCV, LaTeX, GitHub, Google Colab

Contact

Email: varshasingh.aiml@gmail.com

GitHub: <https://github.com/varsha309>